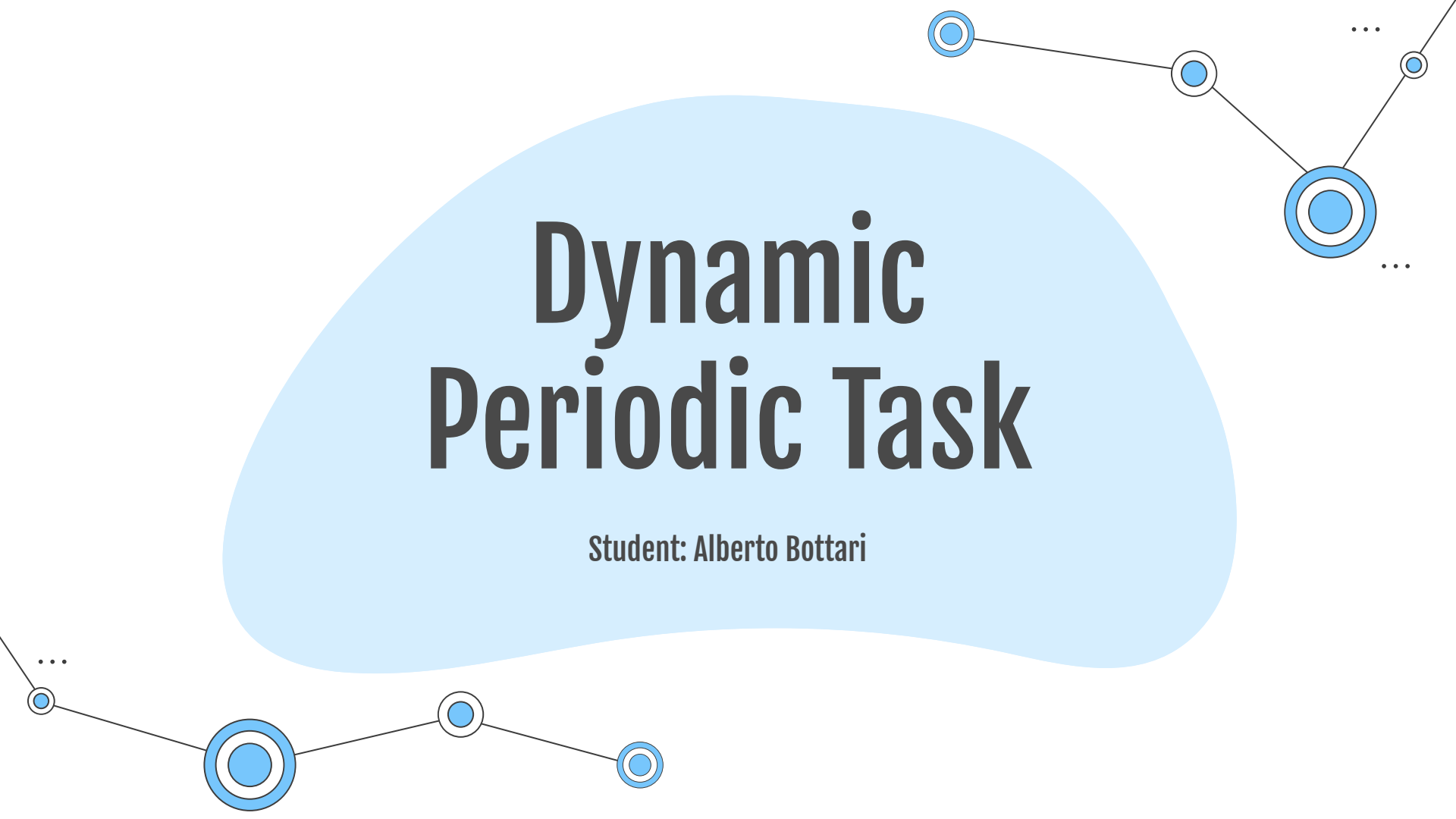




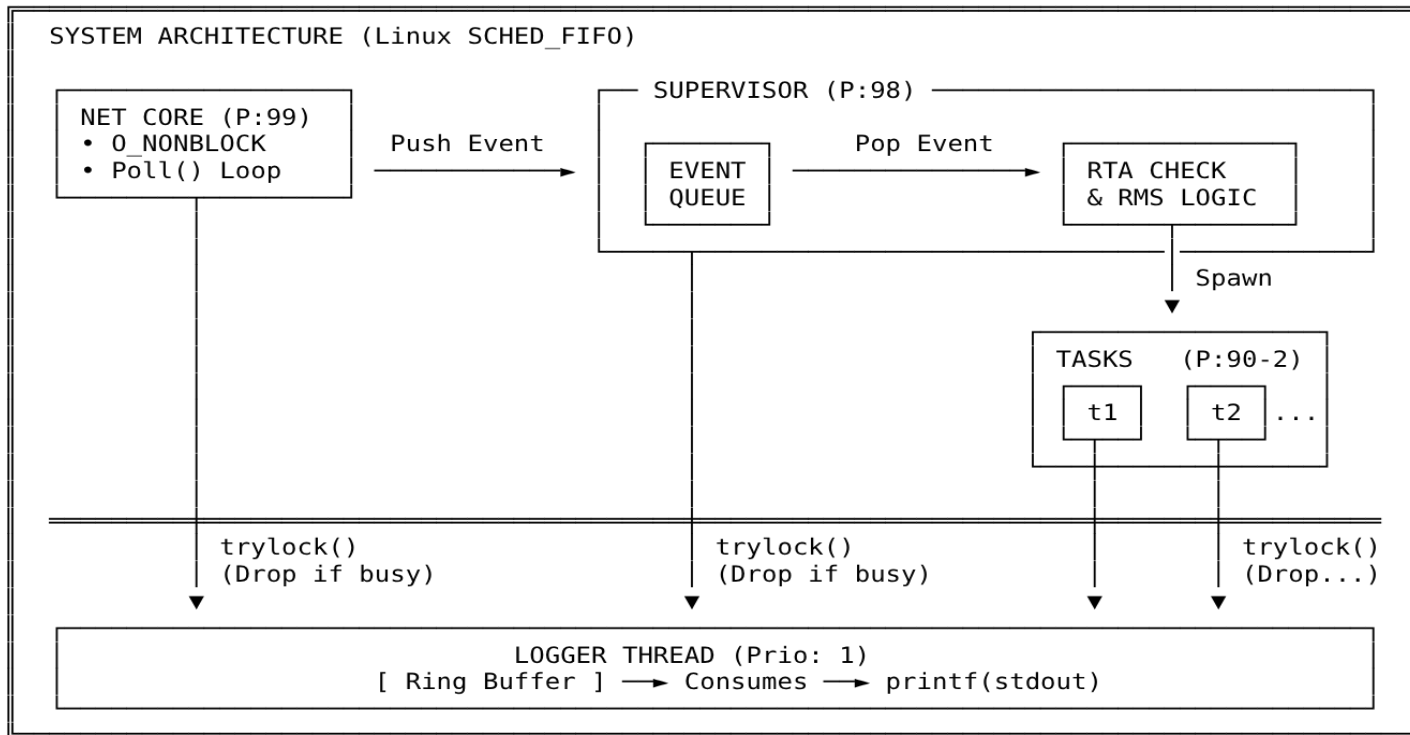
Dynamic Periodic Task

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Architecture

USER (Telnet/NC)

TCP/IP "ACTIVATE t1"



Network Core

01

Mechanism

I/O Multiplexing via `poll()` on a single thread

02

Configuration

Sockets set to `O_NONBLOCK`, so one thread serves more clients

03

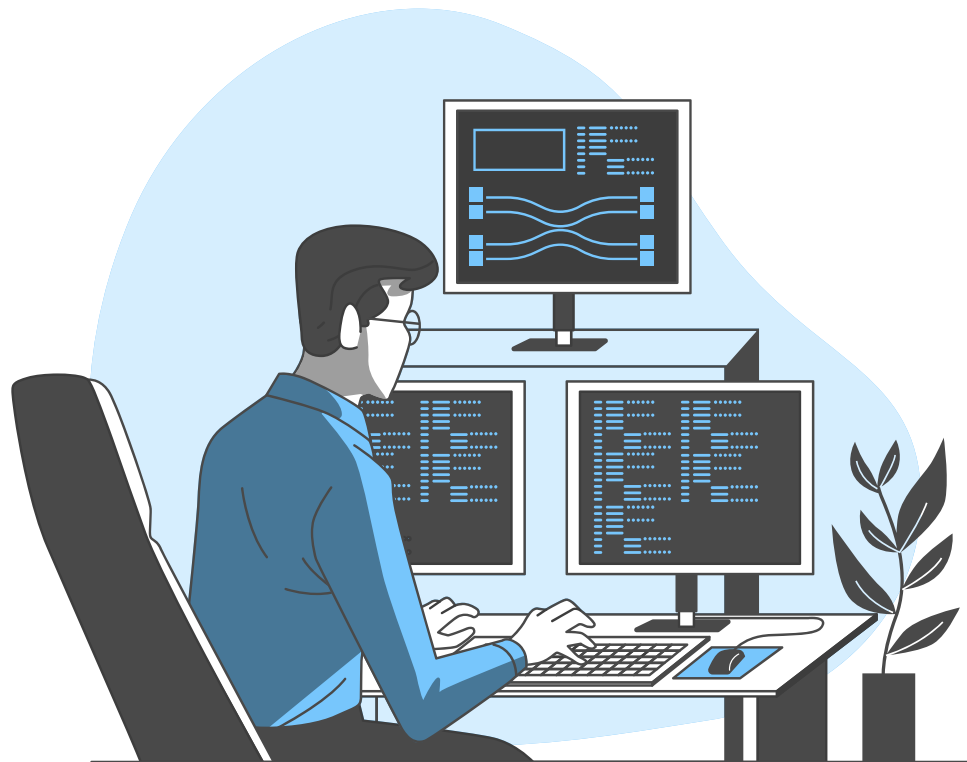
Maximum Priority

Drain sockets fast, avoid client drops

04

Data Integrity

Ring buffers to handle TCP fragmentation



Real-Time Scheduling & RTA



Policy

Rate Monotonic Scheduling (**RMS**): optimal fixed-priority policy where shorter periods receive higher priority.



Admission Control

New tasks are reject if their worst-case response time, under higher-priority interference, exceeds deadline.



Timing Loop

Uses `clock_nanosleep` with `TIMER_ABSTIME` to ensure zero accumulated drift

...

Measures execution time to detect deadline misses and task lateness

What if a deadline is missed?

The Problem

If a task is delayed by seconds, its next activation is in the past.



Task loops immediately to "catch up", starving lower-priority tasks.



The Solution

If `next_activation` $\leq$ `now`, jump forward by N periods.



Maintains the original time grid alignment and recovers system stability instantly.

Why a custom Logger?

Problem

The `printf()` sys-call is thread-safe by using internal locking



if a lower-priority thread holds this lock, a higher-priority thread can block behind it, adding unbounded latency

Solution

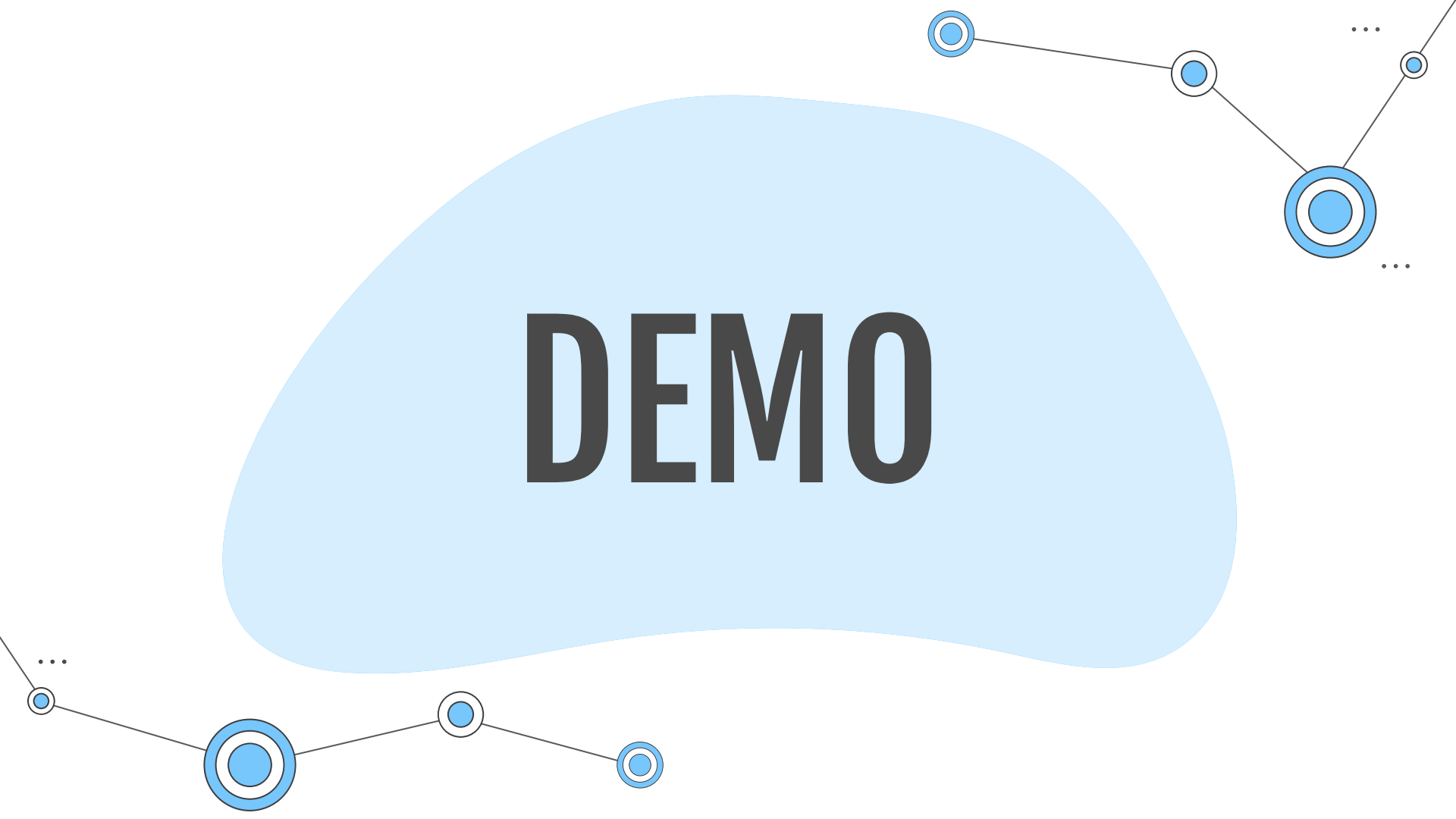
Dedicate a low-priority logger thread that does the blocking `printf()` work.



real-time threads call `pthread_mutex_trylock();`
if contended, the log message is dropped (never block the producer).



DEMO



The image features a light blue, irregular blob shape in the center, containing the word "DEMO" in a bold, black, sans-serif font. Surrounding this central element is a decorative network diagram. It consists of several blue circular nodes, each with a white center and a blue outer ring. These nodes are connected by thin, dark gray lines. The nodes are arranged in a non-linear fashion, with some at the top, some at the bottom, and some on the sides. There are also three sets of three dots (ellipses) indicating that the network continues beyond the visible nodes.